

Hapus Scout™ Enterprise

Multimodal Vision AI Orchard Inspection & Agronomic Evidence Triage Platform for Alphonso Mangoes

1. Delayed Disease Triage & Crop Loss

- Problem: Alphonso mangoes suffer 35-40% annual crop loss from unmonitored pest outbreaks.
- Solution: Sub-6.5s vision AI triage converts worker photos into actionable agronomic briefs.
- ROI & Scale: Replaces ₹1,500/visit fees; enables 1 agronomist to oversee 500+ orchard blocks.

2. Field Worker Language Barrier

- Problem: Over 85% of Konkan orchard workers are native Marathi or Hindi speakers.
- Solution: Multilingual UI (EN, MR, HI) dynamically aligns local terms (Karpa) to AI tensors.
- ROI & Scale: Zero worker training costs; seamless adoption across regional labor pools.

3. Over-Spraying & Export Rejections

- Problem: Farmers spend ₹15k-40k/acre on panic chemical spraying, causing export rejections.
- Solution: Grounded ICAR rules isolate physical observations from unconfirmed hypotheses.
- ROI & Scale: Cuts pesticide spending by 40-60%; protects EU/US premium export compliance.

4. Fragmented Logs & Lack of Audits

- Problem: Informal WhatsApp/paper notes leave farm managers with zero auditable evidence logs.
- Solution: Centralized CaseStore with timestamped audit trails & 1-click agronomist reviews.
- ROI & Scale: Multi-tenant JSON database scales from 2-acre family farms to 1,000+ acre estates.

STRATEGIC EXECUTION & ROI METRICS

- COSTING: 92% Total Cost of Ownership reduction using open 4-bit NF4 vision engine vs. commercial vision APIs.
- PERFORMANCE: Sub-6.5s triage response time processing 1,024 vision patches at ~118ms/tok latency on T4 GPU.
- SCALABILITY: Multi-tenant JSON storage architecture supporting regional multilingual scale across Konkan.

AGRONOMIC GUARDRAIL & REPOSITORY INFORMATION

- Agronomic Guardrail Notice: AI analysis supports triage evidence isolation. Certified agronomist verification mandatory before chemical application.
- Live Repository: <https://github.com/HarshadHindlekar/Hapus-Scout.git>